RegChallenger: A RAG Pipeline for Assessing the Legal Vulnerability of Federal Regulations

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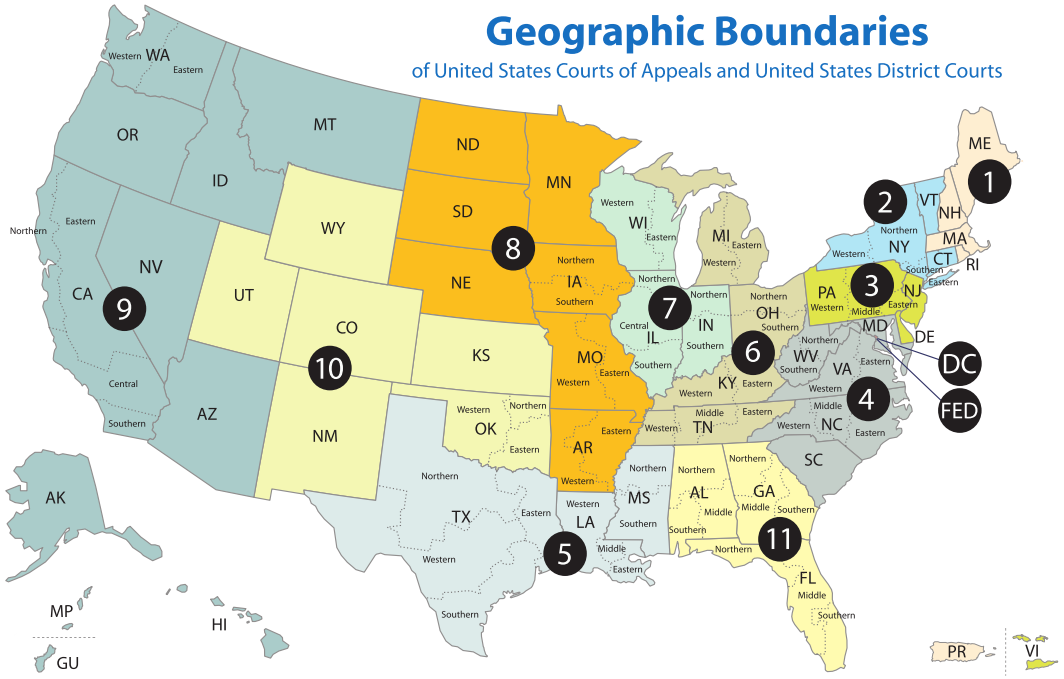
Problem

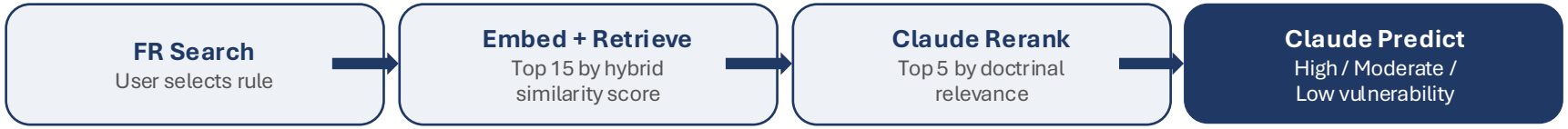
- Agencies regulate nearly every sector of the economy and most aspects of Americans' lives. For example: EPA, FDA, DHS, DOJ, SEC, OSHA, etc.
- Many rules are challenged in federal courts by regulated entities and advocates (companies, trade associations, advocacy groups, etc.)
- *Loper Bright Enterprises v. Raimondo* overturned 40 years of Chevron deference. Courts now interpret statutes anew.
- Litigation is a **time-consuming** and **expensive** process.
- No systematic way to:
  - Assess which regulations are most vulnerable to legal challenge
  - Understand what characteristics make a rule vulnerable
  - Search for historical challenges to similar rules



Solution

Step 1: Build the Reference Corpus

Step 2: RAG Query Flow (for selected rule)



*Hybrid similarity = semantic embeddings (all-mpnet-base-v2, 768-dim) + categorical boosts for agency, doctrine era, and administration. Every prediction cites specific historical analogs (**auditable, not a black box**).*

Tech Stack

Data & Pipeline

- FJC Integrated Database
- CourtListener API
- Federal Register API
- Python, pandas, numpy
- AWS EC2 (pipeline)

AI & Embeddings

- Anthropic Claude API
- claude-sonnet-4
- sentence-transformers
- all-mpnet-base-v2 (768-dim)
- temperature=0 (reproducible)

Interface

- Streamlit
- Custom CSS (GW theme)
- Playfair Display / Source Sans 3 fonts
- python-dotenv